

Chapter 7: Energy of a system

Systems and environments
Kinetic energy
Potential energy
Conservative and non-conservative forces

Work done by a force
Work KE Theorem

Energy Diagrams

1

Introduction to Energy

- ▶ A variety of problems can be solved with Newton's Laws and associated principles.
- ▶ Some problems that could theoretically be solved with Newton's Laws are very difficult in practice.
 - These problems can be made easier with other techniques.
- ▶ The concept of energy is one of the most important topics in science and engineering.
- ▶ Every physical process that occurs in the Universe involves **energy and energy transfers or transformations.**
- ▶ **Energy is not easily defined.**

Introduction

2

Analysis Model

- ▶ The new approach will involve changing from a **particle model to a system model**.
- ▶ In this chapter, systems are introduced along with three ways to store energy in a system.

Introduction

3

Systems

- ▶ A **system** is a small portion of the Universe.
 - We will ignore the details of the rest of the Universe.
- ▶ A critical skill is to **identify** the system.
 - The first step to take in solving a problem
- ▶ A **valid** system:
 - May be a single object or particle
 - May be a collection of objects or particles
 - May be a region of space
 - May vary in size and shape with time

Section 7.1

4

Problem Solving Notes

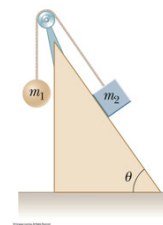
- ▶ The general problem solving approach may be used with an addition to the categorize step.
- ▶ Categorize step of general strategy
 - Identify the need for a system approach
 - Identify the particular system
 - Also identify a system boundary
 - An imaginary surface that divides the Universe into the system and the environment
 - Not necessarily coinciding with a real surface
 - The environment surrounds the system

Section 7.1

5

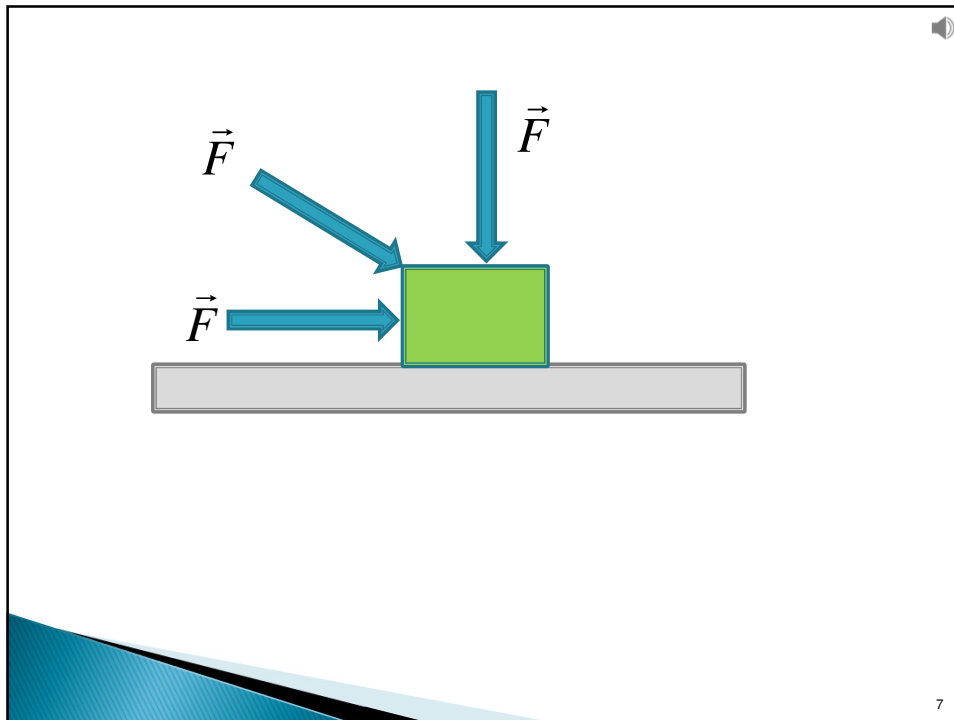
System Example

- ▶ A force applied to an object in empty space
 - System is the object
 - Its surface is the system boundary
 - The force is an influence on the system from its environment that acts across the system boundary.
- ▶ Ball and mass on incline
 - **System** ball, block, cord.
 - **Influence** – gravitational, normal, and frictional forces, force exerted by pulley on cord.
 - Force of cord on ball and block (tension) are **internal forces** – not an influence.



Section 7.1

6



Work

- ▶ The **work**, W , done **on** a system **by** an agent exerting a **constant force** on the system
- ▶ $W = F \Delta r \cos \theta$
 - The displacement is that of the point of application of the force.
 - A force does no work on the object if the force does not move the system through a displacement.
 - The work done by a force on a moving object (system) is zero when the force applied is perpendicular to the displacement of its point of application.
 - The work is a scalar quantity, although the force and the displacement are both vectors

A diagram illustrating the concept of work. It shows a brown rectangular block on a gray horizontal surface. A blue arrow labeled $\vec{F}$ points upwards and to the right, making an angle θ with a horizontal dashed line. Below the block, a horizontal black arrow labeled $\Delta \vec{r}$ indicates the displacement of the block. A small speaker icon is in the top right corner.

Section 7.2

8

Displacement in the Work Equation

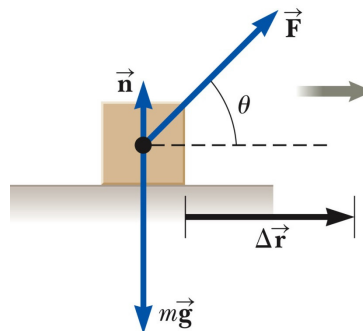
- ▶ If the force is applied to a **rigid object** that can be modeled as a particle, the displacement is the same as that of the particle.
- ▶ For a **deformable system**, the displacement of the object generally is not the same as the displacement associated with the forces applied (e.g. Balloon).

Section 7.2

9

Work Example

- ▶ $W = F \Delta r \cos \theta$
- ▶ The normal force and the gravitational force do no work on the object.
 - $\cos \theta = \cos 90^\circ = 0$
- ▶ The force $\vec{F}$ is the only force that does work on the object.
- ▶ Hold an object at arms length for 3 minutes?



Section 7.2

10

More About Work

- ▶ The **sign** of the work depends on the direction of the force relative to the displacement.
 - Work is **positive** when projection of $\vec{F}$ onto $\Delta\vec{r}$ is in the same direction as the displacement.
 - Work is **negative** when the projection is in the opposite direction.
- ▶ The work done by a force can be calculated, but that force is not necessarily the cause of the displacement.
- ▶ Work is a **scalar quantity**.
- ▶ The unit of work is a **joule (J)**
 - 1 joule = 1 newton · 1 meter = $\text{kg} \cdot \text{m}^2/\text{s}^2$
 - $\text{J} = \text{N} \cdot \text{m}$

Section 7.2

11

Work Is An Energy Transfer

- ▶ If the work done on a system is positive – energy is **transferred to** the system.
- ▶ If the work done on the system is negative – energy is **transferred from** the system.
- ▶ If a system interacts with its environment, this interaction can be described as a **transfer of energy across the system boundary**.
 - This will result in a change in the amount of energy stored in the system.

Section 7.2

12

Example

The gravitational force exerted by the Sun on the Earth holds the Earth in an orbit around the Sun. Let us assume that this orbit is perfectly circular. The work done by this gravitational force during a short time interval in which the Earth moves through a displacement in its orbital path is:

- (a) zero
- (b) Positive
- (c) Negative
- (d) Impossible to determine

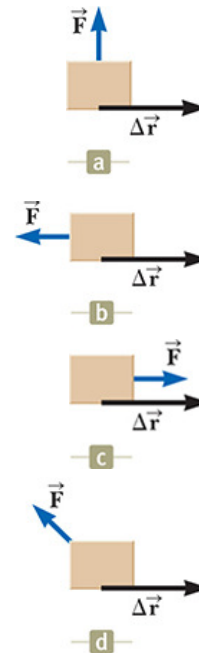
Section 7.2

13

Example

The figure shows four situations in which a force is applied to an object. In all 4 cases, the force has the same magnitude, and the displacement of the object is to the right and of the same magnitude. Which situation :

- (a) has the most positive work done.
- (b) has zero work done.
- (c) has the most negative work done.

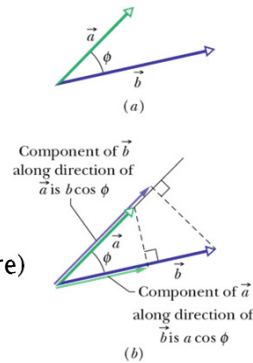


Section 7.2

14

Scalar product

- Two vectors $\vec{a}$ and $\vec{b}$
- Scalar product or dot product
 - $\vec{a} \cdot \vec{b} = ab \cos \phi$
 - a, b magnitudes of the vectors
 - ϕ - angle between the directions of two vectors
- The product of the magnitude of one vector and the scalar component of the other vector along the direction of the first. (see (b) in figure)
- If $\phi = 0^\circ$ then dot product is a max.
- If $\phi = 90^\circ$ then dot product is 0.
- $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$
- $\vec{a} \cdot \vec{b} = a_x b_x + a_y b_y + a_z b_z$



Section 7.3

15

Scalar Product of Two Vectors

- The **scalar product** (or the dot product) of two vectors is written as

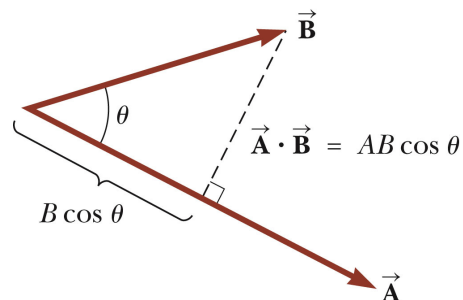
$$\vec{A} \cdot \vec{B} \equiv AB \cos \theta$$

- Applied to work, this means

$$W = F \Delta r \cos \theta = \vec{F} \cdot \Delta \vec{r}$$

- Can also use the components:

$$W = F_x r_x + F_y r_y + F_z r_z$$



Section 7.3

16

Scalar Product

- ▶ Scalar product using unit vector notation

$$\begin{aligned}\vec{a} \cdot \vec{b} &= (a_x \hat{i} + a_y \hat{j} + a_z \hat{k}) \cdot (b_x \hat{i} + b_y \hat{j} + b_z \hat{k}) \\ &= (a_x \hat{i} \cdot b_x \hat{i}) + (a_x \hat{i} \cdot b_y \hat{j}) + (a_x \hat{i} \cdot b_z \hat{k}) \\ &\quad + (a_y \hat{j} \cdot b_x \hat{i}) + (a_y \hat{j} \cdot b_y \hat{j}) + (a_y \hat{j} \cdot b_z \hat{k}) \\ &\quad + (a_z \hat{k} \cdot b_x \hat{i}) + (a_z \hat{k} \cdot b_y \hat{j}) + (a_z \hat{k} \cdot b_z \hat{k}) \\ &= a_x b_x + a_y b_y + a_z b_z\end{aligned}$$

Section 7.3

17

Example

- ▶ Vectors $\vec{C}$ and $\vec{D}$ have magnitudes of 3 units and 4 units, respectively. What is the angle between the directions of the vectors if the dot product equals (i) 0, (ii) 12 units, and (iii) -12 units?

Section 7.3

18

Example 7.3

- ▶ A particle moving in the xy plane undergoes a displacement given by $\Delta\vec{r} = (2.0\hat{i} + 3.0\hat{j})$ m as a constant force $\vec{F} = (5.0\hat{i} + 2.0\hat{j})$ N acts on the particle. Calculate the work done by the force on the particle.

Examples

- ▶ What is the angle ϕ between $\vec{a} = 3.0\hat{i} - 4.0\hat{j}$ and $\vec{b} = -2.0\hat{i} + 3.0\hat{k}$?

Work Done by a Varying Force

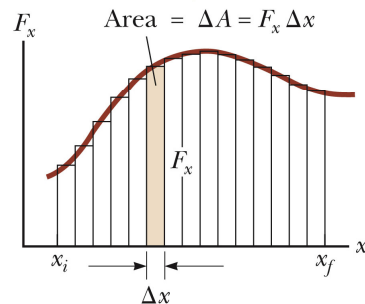
- ▶ To use $W = F\Delta r \cos\theta$, the force must be constant, so the equation cannot be used to calculate the work done by a varying force.

- ▶ Assume that during a very small displacement, Δx , F is constant.

- ▶ For that displacement, $W \sim F\Delta x$

- ▶ For all of the intervals,
- $$W \approx \sum_{x_i}^{x_f} F_x \Delta x$$

The total work done for the displacement from x_i to x_f is approximately equal to the sum of the areas of all the rectangles.



a

Section 7.4

21

Work Done by a Varying Force, cont.

- ▶ As the small displacements approach zero.

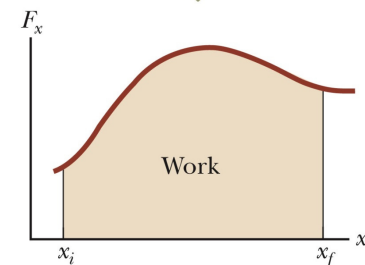
$$\lim_{\Delta x \rightarrow 0} \sum_{x_i}^{x_f} F_x \Delta x = \int_{x_i}^{x_f} F_x dx$$

- ▶ Therefore,

$$W = \int_{x_i}^{x_f} F_x dx$$

- ▶ The work done is equal to the area under the curve between x_i and x_f .

The work done by the component F_x of the varying force as the particle moves from x_i to x_f is *exactly* equal to the area under the curve.



b

Section 7.4

22

Work Done By Multiple Forces

- ▶ If **more than one force acts on a system** and the system can be modeled as a particle, the total work done on the system is the **work done by the net force**.

$$\sum W = W_{\text{ext}} = \int_{x_i}^{x_f} \left(\sum F_x \right) dx$$

- ▶ If the **system cannot be modeled as a particle**, then the total work is equal to the algebraic sum of the work done by the individual forces.

$$\sum W = W_{\text{ext}} = \sum_{\text{forces}} \left(\int \vec{F} \cdot d\vec{r} \right)$$

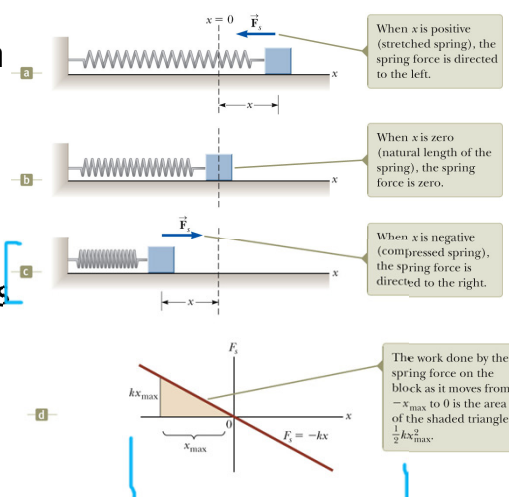
- Remember work is a scalar, so this is the **algebraic sum**.

Section 7.4

23

Work Done By A Spring

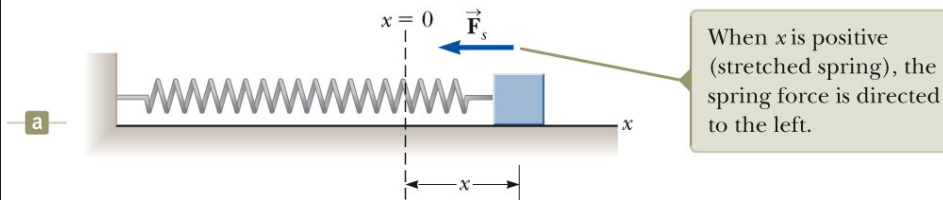
- ▶ A model of a common physical system for which the **force varies** with position is a spring (and block).
- ▶ The **block** is on a horizontal, frictionless surface.



Section 7.4

24

Spring Force (Hooke's Law)



- ▶ The force exerted by the spring is

$$F_s = -kx$$

- x is the **position** of the block with respect to the equilibrium position ($x = 0$).
- k is called the **spring constant** and measures the stiffness of the spring.
- ▶ This is called **Hooke's Law**.

Section 7.4

25

Hooke's Law cont.

$$\vec{F}_s = F_s \hat{i} = -kx \hat{i}$$

- ▶ The force exerted by the spring is always directed **opposite to the displacement** from equilibrium.
- ▶ The spring force is sometimes called the **restoring force**.
- ▶ If the block is released it will **oscillate** back and forth between $-x$ and x .

Section 7.4

26

Work Done by a Spring

Identify the block as the system.

- Calculate the work by spring as the block moves from $x_i = -x_{\max}$

$x_f = 0$.

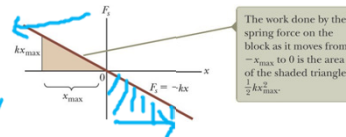
$$W_s = \int \vec{F}_s \cdot d\vec{r} = \int_{x_i}^{x_f} (-kx\hat{i}) \cdot (dx\hat{i})$$

$$= \int_{-x_{\max}}^0 (-kx) dx = \frac{1}{2} kx_{\max}^2$$

- Calculate the work as the block moves from

$x_i = 0$ to $x_f = x_{\max}$. $W = -\frac{1}{2} kx^2$

- The net work done as the block moves from $-x_{\max}$ to $x_{\max}$ is zero.
- Area under the curve.



Section 7.4

27

Work Done by a Spring cont.

- Assume the block undergoes an arbitrary displacement from $x = x_i$ to $x = x_f$.
- The work done by the spring on the block is

$$W_s = \int_{x_i}^{x_f} (-kx) dx = \frac{1}{2} kx_i^2 - \frac{1}{2} kx_f^2$$

- If the motion ends where it begins, $W = 0$ J

Section 7.4

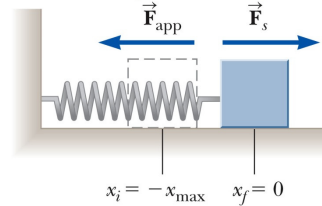
28

Spring with an Applied Force

- Suppose an external agent, F_{app} , moves the spring (controlled).
- The applied force is **equal and opposite** to the spring force.
- $\vec{F}_{app} = F_{app} \hat{i} = -\vec{F}_s = -(-kx\hat{i}) = kx\hat{i}$
- Work done by F_{app} as the block moves from $-x_{max}$ to $x = 0$ is equal to $-\frac{1}{2} kx_{max}^2$
- For any displacement, the work done by the applied force is

$$W_{app} = \int_{x_i}^{x_f} (kx) dx = \frac{1}{2} kx_f^2 - \frac{1}{2} kx_i^2$$

If the process of moving the block is carried out very slowly, then $\vec{F}_{app}$ is equal in magnitude and opposite in direction to $\vec{F}_s$ at all times.



Section 7.4

29

Kinetic Energy

- A possible result of work acting as an influence on a system is that the **system changes its speed**.
- The system could possess **kinetic energy**.
- Kinetic Energy is the energy of a particle due to its motion.
 - $K = \frac{1}{2} mv^2$
 - K is the kinetic energy
 - m is the mass of the particle
 - v is the speed of the particle
- A change in kinetic energy is one possible result of doing work to transfer energy into a system.**

Section 7.5

30

Kinetic Energy, cont

- ▶ Calculating the work:

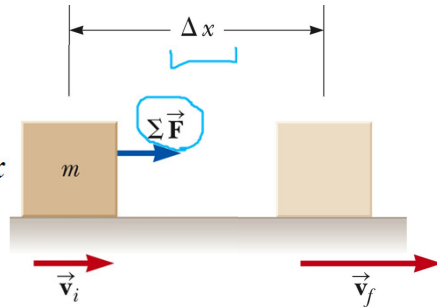
$$W_{ext} = \int_{x_i}^{x_f} \sum F dx = \int_{x_i}^{x_f} ma dx$$

$$W_{ext} = \int_{x_i}^{x_f} m \frac{dv}{dt} dx = \int_{x_i}^{x_f} m \frac{dv}{dx} \frac{dx}{dt} dx$$

$$W_{ext} = \int_{v_i}^{v_f} mv dv$$

$$W_{ext} = \frac{1}{2} mv_f^2 - \frac{1}{2} mv_i^2$$

$$W_{ext} = K_f - K_i = \Delta K$$



Section 7.5

31

Work-Kinetic Energy Theorem

- ▶ The Work-Kinetic Energy Theorem states

$$W_{ext} = K_f - K_i = \Delta K$$

- ▶ When **work is done on** a system **and** the **only change in the system is in its speed**, the net work done on the system equals the change in kinetic energy of the system.

- W is positive $\Rightarrow$ speed of the system increases
- W is negative $\Rightarrow$ speed of the system decreases
- Also valid for changes in rotational speed (later chapters)

Section 7.5

32

Work-Kinetic Energy Theorem

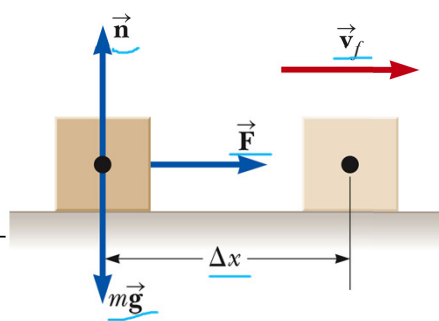
- ▶ The Work-KE theorem **is not valid** when **other changes occur in the system besides speed**, and there are **other interactions** with the environment besides work.
- ▶ The Work-KE theorem relates work to **the speed** of the system, and **NOT to the velocity**.

Section 7.5

33

Work-Kinetic Energy Theorem – Example

- ▶ The block is the system and three external forces act on it.
- ▶ The normal and gravitational forces do no work since they are perpendicular to the direction of the displacement.
- ▶ $W_{ext} = \Delta K = \frac{1}{2} mv_f^2 - 0$
- ▶ The answer could be checked by modeling the block as a particle and using the kinematic equations.



Section 7.5

34

Example

A force $\vec{F} = (3x^2 \text{ N})\hat{i} + (4 \text{ N})\hat{j}$, with x in meters, acts on a particle changing only the kinetic energy of the particle.

- (a) How much work is done on the particle as it moves from coordinates (2 m, 3 m) to (3 m, 0 m)?
- (b) Does the speed of the particle increase, decrease or remain the same?

Section 7.5

35

Example

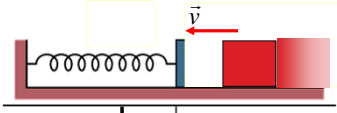
- ▶ A single force acts on a 3.0 kg particle-like object whose position is given by $x = 3.0t - 4.0t^2 + 1.0t^3$, with x in meters and t in seconds. Find the work done on the object by the force from $t = 0$ to $t = 4.0$ s.

Section 7.5

36

Example

In the figure, a canister of mass $m = 0.40 \text{ kg}$, slides across a horizontal frictionless surface with a speed $v = 0.50 \text{ m/s}$. It then runs into and compresses a spring of spring constant $k = 750 \text{ N/m}$. When the canister is momentarily stopped by the spring, by what distance is the spring compressed?



- $m = 0.40 \text{ kg}$
- $v_i = 0.50 \text{ m/s}$
- $v_f = 0 \text{ m/s}$
- $k = 750 \text{ N/m}$

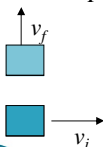
Section 7.5

37

Example

The only force acting on a 2.0 kg canister that is moving in an xy plane has a magnitude of 5.0 N . The canister initially has a velocity of 4.0 m/s in the positive x direction and some time later has a velocity of 6.0 m/s in the positive y direction. How much work is done on the canister by the 5.0 N force during this time?

- $v_i = 4.0 \text{ m/s}$
- $v_f = 6.0 \text{ m/s}$
- $m = 2.0 \text{ kg}$
- $F = 5.0 \text{ N}$



Section 7.5

38

Example

A luge and its rider, with a total mass of 85 kg, emerge from a downhill track onto a horizontal straight track with an initial speed of 37 m/s. If a force slows them to a stop at a constant rate of 2.0 m/s^2

- What is the magnitude F of the net force required?
- What distance d do they travel while slowing down, and
- What work W is done on them (luge and rider) by the force?

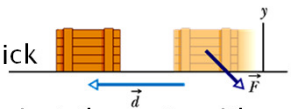
- $m = 85 \text{ kg}$
- $v_i = 37 \text{ m/s}$
- $v_f = 0 \text{ m/s}$
- $a = 2.0 \text{ m/s}^2$

Section 7.5

39

Example

During a storm, a crate is sliding across a slick oily parking lot through a displacement $\vec{d} = (-3.0 \text{ m})\hat{i}$ while a steady wind pushes against the crate with a force $\vec{F} = (2.0 \text{ N})\hat{i} + (-6.0 \text{ N})\hat{j}$. The situation and coordinate axes are shown in the Figure.



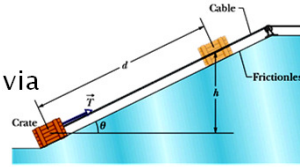
- How much work does this force do on the crate during the displacement?
- If the crate has a kinetic energy of 10 J at the beginning, what is its kinetic energy at the end of the displacement?

Section 7.5

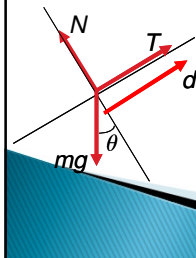
40

Problem

A 15.0 kg crate, initially at rest, is pulled, via a cable, a distance $d = 5.70$ m up a frictionless ramp to a height $h = 2.50$ m, where it stops.



- (a) How much work is done on the crate by the gravitational force during the lift?
(b) How much work is done on the crate by the force T during the lift?
System?



Section 7.5

41

Potential Energy

- ▶ Potential energy is energy (!) determined by the configuration of a system in which the components of the system interact by forces.
 - The forces are internal to the system.
 - PE can be associated with only specific types of forces acting between members of a system.
 - E.g. Book–Earth system, spring–block system.

Section 7.6

42

Gravitational Potential Energy

- ▶ Consider the system of **the Earth and the book**.
- ▶ Do work on the book by **lifting it slowly** (no acceleration) through a vertical displacement

$$\Delta \vec{r} = (y_f - y_i) \hat{j}$$

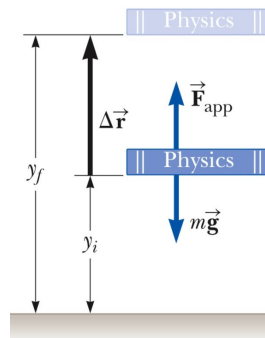
- ▶ The **work done** on the system must appear as an **increase in the energy of the system**.

$$W_{ext} = (\vec{F}_{app}) \cdot \Delta \vec{r}$$

$$W_{ext} = (mg \hat{j}) \cdot [(y_f - y_i) \hat{j}]$$

$$= mgy_f - mgy_i$$

The work done by the agent on the book-Earth system is $mgy_f - mgy_i$.



Section 7.6

43

Gravitational Potential Energy _{cont}

- ▶ The energy storage mechanism is called **potential energy**.
- ▶ **Gravitational potential energy** is the **energy associated with an object at a given location** above the surface of the Earth.

Section 7.6

44

Gravitational Potential Energy, final

- ▶ The quantity mgy is identified as the **gravitational potential energy, U_g** .
 - $U_g = mgy$
- ▶ Units are joules (J), and U_g is a scalar
- ▶ g is approx. constant: 9.8 m/s^{-2}
- ▶ Work may change the gravitational potential energy of the system.
 - $W_{\text{ext}} = \Delta U_g$
- ▶ Potential energy is always associated with a system of two or more interacting objects.

Section 7.6

45

Gravitational Potential Energy, Problem Solving

- ▶ The gravitational potential energy depends only on the vertical height of the object above Earth's surface.
- ▶ You must **choose a reference configuration** for which the gravitational potential energy is set equal to some **reference value, normally zero**.
- ▶ **The surface of the Earth is a convenient zero gravitational potential energy configuration.**
- ▶ See Example 7.8 in textbook.

Section 7.6

46

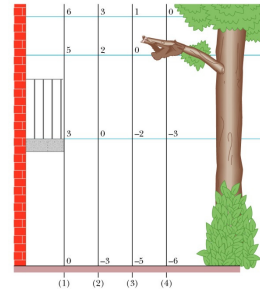
Example

A 2.0 kg sloth hangs 5.0 m above the ground.

(a) What is the gravitational potential energy U of the sloth–Earth system if we take the reference point $y = 0$ to be

- At the ground,
- At the balcony floor (3 m above the ground)
- At the branch
- 1.0 m above the limb?

(b) The sloth drops to the ground. For each of the reference points, what is the ΔU in the potential energy of the Sloth–Earth system.



47

Elastic Potential Energy

- ▶ **Elastic Potential Energy** is associated with a spring.
- ▶ The force the spring exerts (on a block, for example) is $F_s = -kx$
- ▶ The work done by an external applied force on a spring–block system is
 - $W_{\text{ext}} = \frac{1}{2} kx_f^2 - \frac{1}{2} kx_i^2$
 - The work is equal to the difference between the initial and final values of an expression related to the configuration of the system.
- ▶ The expression for elastic potential energy:

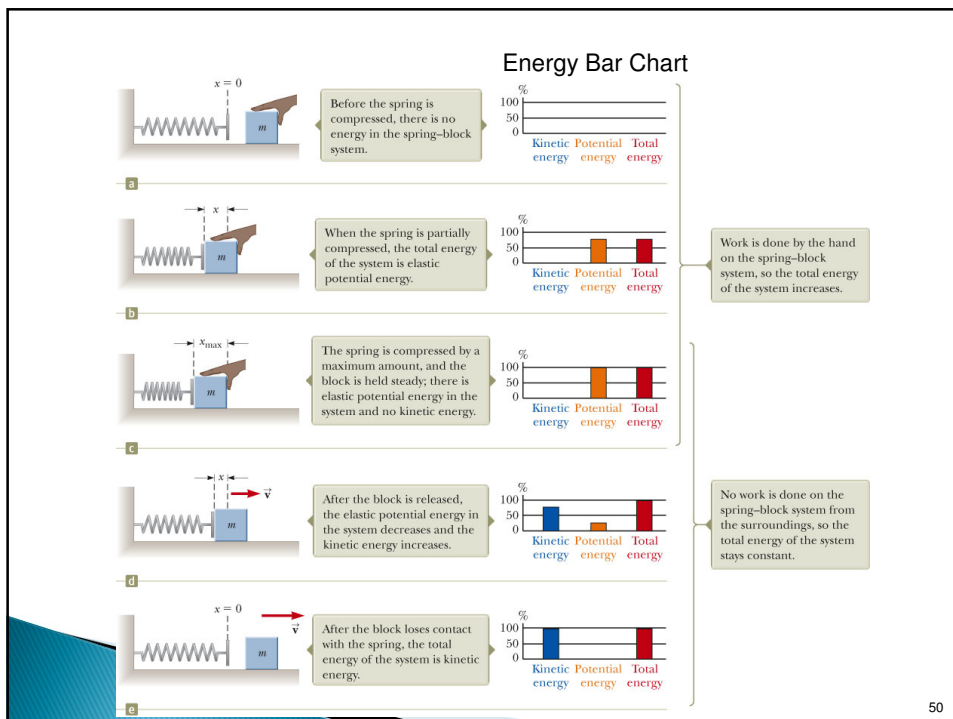
$$U_s \equiv \frac{1}{2} kx^2$$
- ▶ $W_{\text{ext}} = \Delta U_s$

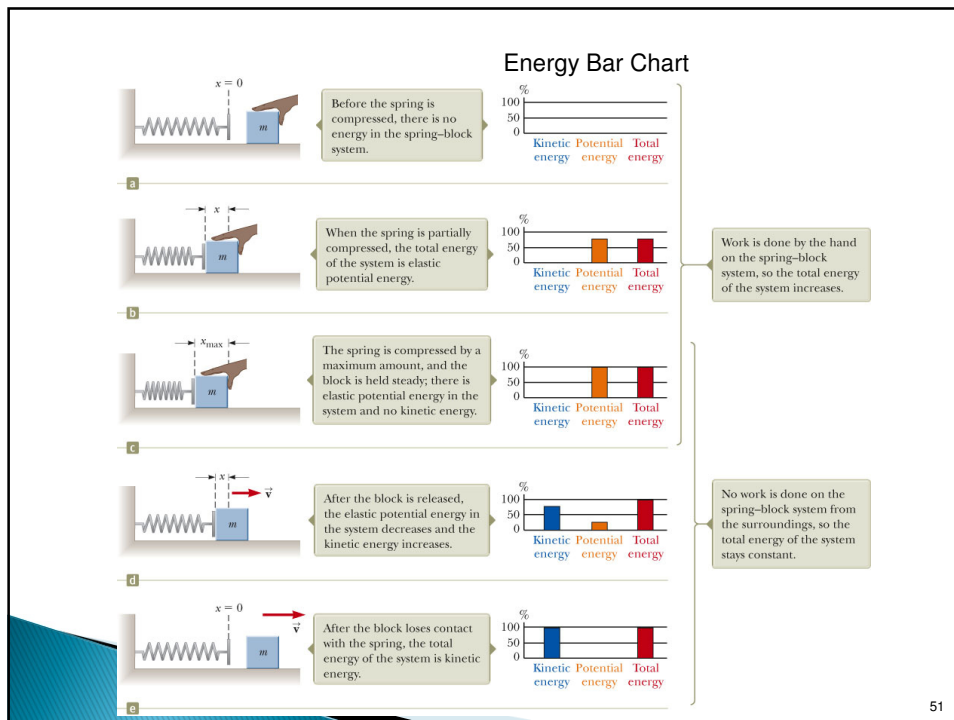
Section 7.6

48

Elastic Potential Energy cont.

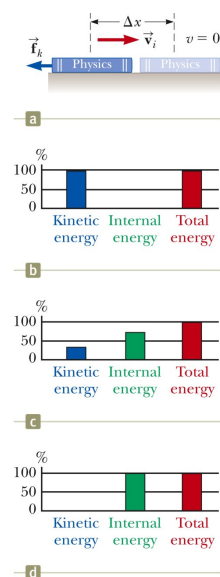
- ▶ The elastic potential energy can be thought of as the **energy stored in the deformed spring**.
- ▶ The stored PE can be converted into KE.
- ▶ Example Fig 7.17





Internal Energy

- ▶ The energy associated with an object's temperature is called its *internal energy*, E_{int} .
- ▶ Example: Book slides across the surface.
- ▶ In this example, **the surface is the system**.
- ▶ The **friction** does work and **increases** the internal energy of the surface.
- ▶ **When the book stops**, all of its kinetic energy has been transformed to internal energy.
- ▶ The **total energy remains the same**.



Section 7.7

Conservative Forces

- ▶ Conservative forces have these two properties:
- ▶ The **work done** by a conservative force on a particle moving between any two points is **independent of the path taken by the particle**.
- ▶ The **work done** by a conservative force on a particle **moving through any closed path is zero**.
 - A closed path is one in which the beginning and ending points are the same.
- ▶ Examples of conservative forces:
 - Gravity
 - Spring force

Section 7.7

53

Conservative Forces, cont

- ▶ We can associate **a potential energy for a system** with any **conservative force acting between members of the system**.
 - In general: $W_{int} = -\Delta U$
 - W_{int} is used as a reminder that the work is done by one member of the system on another member and is internal to the system.
 - **Positive work done by an outside agent** on a system causes an increase in the potential energy of the system.
 - **Work done** on a component of a system **by a conservative force internal** to an isolated system causes a decrease in the potential energy of the system.

Section 7.7

54

Non-conservative Forces

- ▶ A non-conservative force **does not satisfy the conditions of conservative forces.**
- ▶ Non-conservative forces **acting in a system cause a change in the mechanical energy** of the system.
- ▶ $E_{mech} = K + U$
 - K includes the kinetic energy of all moving members of the system.
 - U includes all types of potential energy in the system.
 - Does not include E_{int}

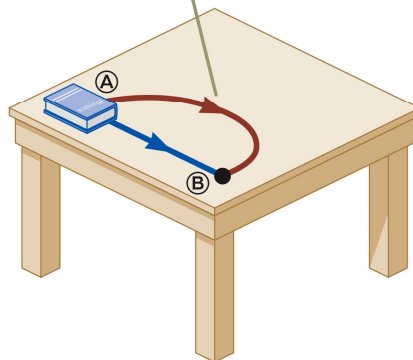
Section 7.7

55

Non-conservative Forces, cont.

- ▶ Book is displaced from A to B on a surface with friction.
- ▶ The work done against friction is greater along the brown path than along the blue path.
- ▶ Because the work done depends on the path, friction is a non-conservative force.

The work done in moving the book is greater along the brown path than along the blue path.



Conservative Forces and Potential Energy

- ▶ Define a potential energy function, U , such that the **work done by a conservative force equals the decrease in the potential energy** of the system.
- ▶ The work done by such a force, F , is

$$W_{\text{int}} = \int_{x_i}^{x_f} F_x dx = -\Delta U$$

- ΔU is negative when F and x are in the same direction
- E.g. When an object falls back down to Earth.

Section 7.8

57

Conservative Forces and Potential Energy

- ▶ The conservative force is related to the potential energy function through:

$$F_x = -\frac{dU}{dx}$$

- ▶ The x component of a conservative force acting on an object within a system equals the negative derivative of the potential energy of the system with respect to Δx .
 - E.g 1. block moving up a slope: conservative force is gravity, potential energy is $(-mg\Delta y)$
 - E.g.2. Block spring system: conservative force is spring force, potential energy is $(\frac{1}{2}kx^2)$
 - This can be extended to three dimensions

Section 7.8

58

Conservative Forces and Potential Energy – Check

- ▶ Look at the case of a deformed spring:

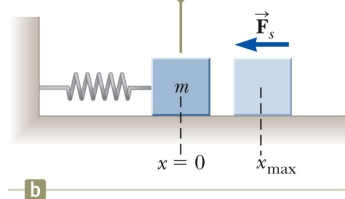
$$F_s = -\frac{dU_s}{dx} = -\frac{d}{dx}\left(\frac{1}{2}kx^2\right) = -kx$$

- This is Hooke's Law and confirms the equation for U
- ▶ U is an important function because a conservative force can be derived from it.

59

Energy Diagrams and Equilibrium

The restoring force exerted by the spring always acts toward $x = 0$, the position of stable equilibrium.



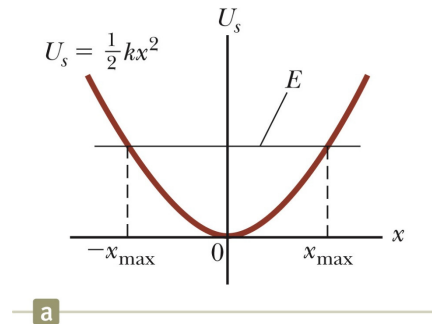
- ▶ Motion in a system can be observed in terms of a graph of its position and energy.
- ▶ In a spring-mass system example, the block oscillates between the turning points, $x = \pm x_{\max}$.
- ▶ The block will always accelerate back toward $x = 0$.

Section 7.9

60

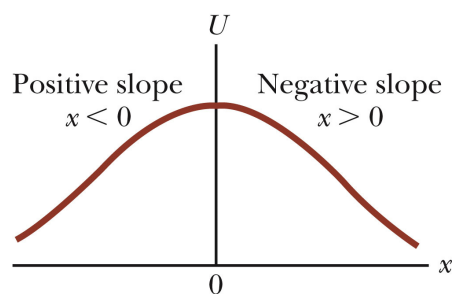
Energy Diagrams and Stable Equilibrium

- ▶ The $x = 0$ position is one of **stable equilibrium**.
 - Any movement away from this position results in a force directed back toward $x = 0$.
- ▶ Configurations of stable equilibrium correspond to those for which $U(x)$ is a **minimum**.
- ▶ $x = x_{\max}$ and $x = -x_{\max}$ are called the **turning points**.



Energy Diagrams and Unstable Equilibrium

- ▶ $F_x = 0$ at $x = 0$, so the particle is in **equilibrium**.
- ▶ For any other value of x , the particle moves away from the equilibrium position.
- ▶ This is an example of **unstable equilibrium**.
- ▶ Configurations of unstable equilibrium correspond to those for which $U(x)$ is a **maximum**.



Neutral Equilibrium

- ▶ **Neutral equilibrium** occurs in a configuration when U is constant over some region.
- ▶ A small displacement from a position in this region will produce neither restoring nor disrupting forces.
- ▶ E.g. ball lying in the middle of a table.

Section 7.9

63

Potential Energy in Molecules

- ▶ There is potential energy associated with the force between two neutral atoms in a molecule which can be modeled by the *Lennard-Jones* function.

$$U(x) = 4\epsilon \left[\left(\frac{\sigma}{x} \right)^{12} - \left(\frac{\sigma}{x} \right)^6 \right]$$

- ▶ Find the minimum of the function (take the derivative and set it equal to 0) to find the separation for stable equilibrium.
- ▶ The graph of the Lennard-Jones function shows the most likely separation between the atoms in the molecule (at minimum energy).

